		Constant velocity, for F = resistive force = 25 N
8	**D**	Centripetal force $F_c = mr\omega^2$ is provided by the friction between the object and the turntable.

[Turn over

		Before slipping, centripetal force of Q > P due to larger *r*. Option A & B: Incorrect. Friction of Q > P. Option C: Incorrect. Q has the same angular velocity as P. Option D: Correct. Friction provided by turntable has a maximum value. Required centripetal force of Q will exceed maximum friction first due to larger *r*.
9	**C**	Horizontal: $T\ sin\ sin\ \theta\ = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \ldots (1)$